

COMPRESS

Describe what stays and what goes

01 / THE SAS EXAMPLE

```
data test;
  infile datalines truncover;
  input text $char20.;
  no_spaces = compress(text);
  no_digits = compress(text, '0123456789');
  only_letters = compress(text, '', 'ka');
  datalines;
ABC 123 XYZ
Phone: 555-1234
ID-2023-Q4
;
run;
proc print data=test noobs;
run;
```

ATTACH SOURCE OR PASTE IT AFTER THE PROMPT. START FRESH EACH TIME.

WHAT YOU WILL PRACTICE

Replace vague cleaning instructions with concrete character rules.

Three strings illustrate three rules: remove spaces, remove digits, or keep letters. For these ASCII inputs, ka keeps alphabetic characters. Specify what cleaning should preserve.

EXPECTED RESULT

Row	no_spaces	no_digits	only_letters
1	"ABC123XYZ"	"ABC XYZ"	"ABCXYZ"
2	"Phone:555-1234"	"Phone: - "	"Phone"
3	"ID-2023-Q4"	"ID- -Q"	"IDQ"

Quotes mark string boundaries and are not data. Stored trailing padding is omitted. Preserve both internal spaces in the first no_digits result.

Expected values checked in Python; SAS execution pending.

Prompt A: leave choices open

High degrees of freedom / Notice the decisions the model makes for you.

COPY THIS PROMPT + THE SAS EXAMPLE

Convert this SAS code to Python.
Show code in this response. Do not create or run files, or read existing Python files.

LOOK AT THE RESPONSE

Which library and method did it choose? Did it preserve the expected behavior? Compare the result with page 1 before accepting it.

TRY IT / THREE FRESH ATTEMPTS

Run A three times with the same SAS example and setup. Save each first response before asking follow-up questions. Record agreement as well as differences.

Bedrock A/B trials for this example are pending.

WHY CHOICES REMAIN OPEN

- 1 A usable starting point**
The source shows the operations and strings. A short prompt may already produce a correct translation.
- 2 A shorthand to interpret**
The model must interpret ka correctly. Keeping letters and digits together would change the requested result.
- 3 Several valid routes**
Replacement, regular expressions, and filtering can all work here. Different implementation does not automatically mean incorrect behavior.

ANTI-PATTERN TO AVOID

Anti-pattern: "Clean up the strings." That leaves the model to choose what disappears. Specify allowed characters and provide a before-and-after pair for each rule.

Prompt B: make the choices explicit

Low degrees of freedom / State the behavior, then ask for a check.

COPY THIS PROMPT + THE SAME SAS EXAMPLE

```
Convert this SAS code to Python using pandas.
Show code in this response. Do not create or run files, or read
existing Python files.

- Recreate the three text values in their original order.
- These inputs contain ASCII characters only. Ignore SAS trailing
  storage padding when comparing values; preserve internal spaces.
- no_spaces: remove literal spaces with
  .str.replace(' ', '', regex=False).
- no_digits: remove 0-9 with .str.replace('[0-9]', '',
  regex=True).
- only_letters: ka keeps letters; use
  .str.replace('[^A-Za-z]', '', regex=True).
- Include assertions for these three complete result columns:
  no_spaces: ['ABC123XYZ', 'Phone:555-1234', 'ID-2023-Q4']
  no_digits: ['ABC XYZ', 'Phone: -', 'ID--Q']
  only_letters: ['ABCXYZ', 'Phone', 'IDQ']
- Return one complete script. Compute results from text;
  do not hard-code the result columns.
Briefly explain the difference between removing and keeping.
```

WHY THESE DETAILS HELP

- 1 Separate the rules**
Each column has its own instruction. Removing digits does not permit removing punctuation or internal spaces.
- 2 Translate the shorthand**
Explaining ka as keep letters makes the intended behavior explicit. ASCII defines the scope of this exercise.
- 3 Check the small details**
Quoted expected values make internal spaces and punctuation visible parts of the check.

CHECK IT YOURSELF

Run the script separately. Check every string, including retained punctuation. Confirm two internal spaces in the first no_digits value. Neat appearance cannot establish that required characters survived.

When asking for cleaning, specify exactly which characters to keep or remove.